

PALLAVI KOCHAR

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EDUCATION

University of Illinois Urbana-Champaign (UIUC) (Urbana-Champaign, IL) Jan 2025 - May 2026

- Master of Science in Finance (Specialization: Asset Management, Concentration: Data Analytics, Quantitative Finance)
- Research Assistant - DeFi Crypto Asset Pricing, Course Assistant - Financial Economics, MSF Program Ambassador
- Coursework: Machine Learning, Applied Portfolio Management, Advanced Financial Derivatives, Big Data Analytics

Indian Institute of Technology (IIT), Bombay (Mumbai, India) Jun 2018 - May 2022

- Bachelor of Technology (BTech) in Chemical Engineering (Minor: Industrial Engineering and Operations Research)
- Coursework: Numerical Analysis, Calculus, Economics, Probability and Stochastic Processes, Differential Equations

PROFESSIONAL EXPERIENCE

Kotak Securities Limited | Backend Engineer (Mumbai, India) Jun 2022 - Dec 2024

- Spearheaded the end-to-end design and implementation of a fully automated SIP order auto-debit system for Smallcase SIP baskets, processing 10K+ monthly transactions with 99.9% real-time accuracy, leveraging C#, SSIS and SQL Server
- Built scalable AWS-based data pipelines for Salesforce CRM, accelerating customer insight access and improving analytics workflow efficiency by 30%
- Automated multi-stock price update process and saved 200 IT hours monthly, strengthening intraday analytics reliability
- Streamlined and optimized 5+ micro-services for corporate actions, gain/loss processes, and data verification within the Portfolio Tracker, reducing IT weekly workload by 50+ hours and enhancing stock performance analysis efficiency

FAlpha.ai | Data Product Engineer Intern (New York City, USA) Apr 2026 - May 2026

- Built an Explainable AI layer over prediction models, translating opaque sequence predictions into per-feature attributions
- Benchmarked SHAP against alternative attribution methods (LIME, integrated gradients), recommending the approach that balanced fidelity and runtime for production deployment

FlexiEle | AI / ML Internship (Gurgaon, India) May 2021 - Jul 2021

- Designed an AI chat-bot using NLP techniques to conduct initial screenings, reducing manual recruiter workload by 60%
- Trained ML algorithms to assess responses, improving prediction accuracy by 25% compared to rule-based methods

KEY PROJECTS

Multi-Agent trading System (Course: Quantamental Investment) Apr 2026 - May 2026

- Architected a multi-agent equity-trading system for Permian E&P names enforcing a deterministic Python numerical core with GPT-4o-mini agents confined to qualitative roles, eliminating the LLM-calibration failure mode
- Backtested a satellite-anchored pre-earnings strategy across 10 Permian E&Ps; generated 18 trades with 83% hit rate, +13.5% return on \$1M, and 0.35 per-trade Sharpe, with pre-registered ablations validating signal source

Structured Product Valuation: UBS Auto-Callable Securities (Course: Financial Derivatives) Apr 2026 - May 2026

- Priced a 2-year worst-of auto-callable note on AAPL/AMZN/GOOGL via Python Monte Carlo, simulating correlated GBM paths via Cholesky decomposition with Bloomberg-sourced ATM vols, OIS discounting, and continuous dividend drift
- Modeled path-dependent payoff (memory coupon, quarterly autocall, 50% barrier), priced at \$978/\$1,000 face, and quantified volatility and correlation sensitivities to isolate the embedded short worst-of put

Behavioral Analysis of Credit Card Advertising (Research Project) Jan 2026 - May 2026

- Engineered a data pipeline to download and structure Competiscan credit card advertising data, building the foundation for behavioral analysis of feature salience across thousands of ad creatives
- Designed and ran experiments to test the impact of ad design and feature highlighting on credit card selection, using quantitative analysis to identify which product attributes most strongly drive user preference and confidence

GARCH Volatility and VaR Modeling (Course: Financial Risk Management) Mar 2026 - Apr 2026

- Estimated GARCH(1,1) on SPY returns via custom Maximum Likelihood in R, benchmarking against tseries and fGarch package estimates, and forecasted 1-day and 21-day annualized volatility
- Computed 1-day VaR and Expected Shortfall at 95% confidence, benchmarking GARCH-based estimates against historical simulation and delta-normal VaR to compare tail-risk coverage across approaches

SKILLS AND AWARDS

- **Technical:** Python (Pandas, NumPy, scikit-learn, Tensorflow, Keras), SQL, R, ETL, RDBMS, CI/CD, AWS, Azure, Bloomberg
- **Finance & Quant:** Derivative Pricing (Black Scholes, Binomial Trees), Monte Carlo Simulation, Hedging Strategies, DCF
- **Machine Learning:** Regression, Clustering, Time Series Forecasting, Neural Networks (LSTM, FinBERT), Explainable AI
- **AI Tools:** Claude, Anthropic Cowork, GitHub Copilot, Cline, AntiGravity, multi-agent orchestration, Codex, MCP
- **Certifications:** [Finance and Quantitative Modeling](#), [Machine Learning](#), [IBM Data Science](#), [LangChain](#)
- **Awards:** Advanced 3 designations at Kotak Securities in 2 years · IT Champion Award FY 2022-23 · Top 0.1% JEE-Mains